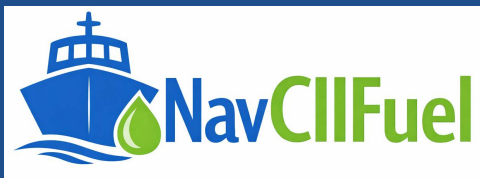


MARITIME COMPLIANCE TOOLS

USER GUIDE

2-Stroke Engine —SFOC Correction Calculator — Web Version



Specific Fuel Oil Consumption Correction · Web Version

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1. Introduction

1.1 Purpose of this guide

This guide explains how to use the 2-Stroke Engine — ISO 3046 SFOC Correction Calculator, a browser-based tool for correcting measured Specific Fuel Oil Consumption (SFOC) to ISO 3046 reference conditions. It walks through every input field, the engine load table, the SFC calculator for PME, the results and comparison charts, and the built-in on-screen guide, using screenshots taken directly from the tool.

The guide is written for engine room officers, technical superintendents, and performance engineers who need to enter shop or sea-trial test data, apply ISO ambient corrections, and produce a corrected SFOC figure for reporting or trending purposes.

1.2 What the tool does

- Corrects measured SFOC to ISO 3046 standard reference conditions (ambient temperature, barometric pressure, charge-air coolant temperature and fuel LCV).
- Applies the linear correction method (per the vessel's Technical File) to calculate temperature, pressure, coolant and LCV correction percentages at each engine load point.
- Supports any number of engine load points (default 25%, 50%, 75%, 100% MCR), with load columns that can be added, edited or removed.
- Displays a summary of ISO-corrected SFOC results and plots measured vs. ISO-corrected SFOC on interactive comparison charts.
- Includes an optional SFC Calculator for PME, which interpolates the ISO-corrected SFC at the vessel's actual propulsion power point.
- Provides a built-in, on-screen User Guide panel (the ? / User Guide button) describing every field, formula and colour code used in the tool.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have the engine's shop trial or sea trial report to hand, including measured SFOC, ambient temperature, barometric pressure, charge-air coolant temperature and fuel LCV at each load point.
- Use a current version of Chrome, Edge, or Firefox for the best experience.

2. Interface Overview

The calculator opens with a fixed top bar, an ISO Reference Conditions strip, the Engine Load Controls, the Engine Data table, and — once Calculate is run — the Results Summary, SFOC Comparison Charts and the optional SFC Calculator for PME section.



Figure 1 — Top bar with Back, Home and User Guide controls.

Control	Function
◀ Back	Returns to the previous page in the browser.
🏠 Home	Returns to the tool's home / landing page.
User Guide (?)	Opens the built-in on-screen guide describing every field, formula and colour code in the tool.

The page body is organised top to bottom into five functional blocks, described in detail in the following sections:

Block	Purpose
ISO Reference Conditions	Fixed reference values (T_r , p_r , T_{cr} , reference LCV) that all corrections are measured against.
Engine Load Controls	Add, edit or remove the engine load percentage columns used across the table.
Engine Data Table	Enter measured SFOC and ambient test data; view calculated correction steps and the ISO-corrected result.
Results Summary & Charts	Appears after Calculate — shows mean ISO-corrected SFOC and a measured-vs-corrected comparison chart.
SFC Calculator for PME	Optional section — estimates the ISO-corrected SFC at the vessel's actual propulsion power.

3. Quick Start Workflow

Follow these seven steps for a typical single-engine SFOC correction:

Step	What to do
1	Open the calculator HTML file in your browser and confirm the ISO Reference Conditions strip, load controls and engine table all load correctly.
2	In the ISO Reference Conditions strip, confirm or edit the reference ambient temperature, pressure, coolant temperature and fuel LCV for your Technical File.
3	In Engine Load Controls, confirm the load points (default 25 / 50 / 75 / 100% MCR) or use + Add Load / × Remove to match your test record.
4	In the Engine Data table, enter the Engine No. and Order No., then enter SFOC Measured, Tx, Px, Tcx and G (fuel LCV) for each load point.
5	Click Calculate to run the ISO correction. Review the correction percentages, the SFOC ISO Corrected row, and the Results Summary.
6	Review the SFOC Comparison chart (measured vs. ISO corrected) for each load point.
7	If required, switch on the SFC Calculator for PME, choose whether MCRI _{lim} is available, and enter PME / MCR to obtain the corrected SFC at the vessel's actual operating point.

Tip

- Use Reset defaults at any time to restore the tool's standard sample values for the current engine.
- The ? / User Guide button in the top bar opens a full on-screen reference for every field and formula used by the tool.

4. ISO Reference Conditions

This strip at the top of the page defines the ISO 3046 baseline that every correction is measured against. Values are pre-filled with the ISO 3046 standard defaults but can be edited to match the vessel's approved Technical File.

ISO 3046 REFERENCE

T_a AMBIENT

25

°C

P_b PRESSURE

1000

mbar

T_{cr} COOLANT

25

°C

H REF LCV

42700

kJ/kg

Figure 2 — ISO Reference Conditions strip with default ISO 3046 values.

Field	Description	Default
Tr — ambient	ISO reference ambient (charge-air) temperature used in the temperature correction.	25 °C
pr — pressure	ISO reference barometric pressure used in the pressure correction.	1000 mbar
Tcr — coolant	ISO reference charge-air coolant (jacket water) temperature.	25 °C
H ref LCV	ISO reference fuel lower calorific value used in the LCV correction.	42,700 kJ/kg
Tip - Changing any reference value automatically re-runs the correction calculation across every load point (recalcAll). - Only change these reference values if the vessel's Technical File specifies different ISO baseline figures.		

5. Engine Load Controls

The load chips control which engine load percentages appear as columns across the Engine Data table, the Results Summary and the comparison charts. The tool ships with four standard MCR load points.



Figure 3 — Load chips (25%, 50%, 75%, 100%) with + Add Load control.

Control	Function
Load chip value	Click into a chip and type a new load percentage (1–100) to relabel that column.
× (remove)	Deletes that load column. At least one load column must remain.
+ Add Load	Appends the next available standard load point (25 / 50 / 75 / 100%) as a new column.
Tip • The tool offers up to the four standard MCR points (25%, 50%, 75%, 100%). Once all four are in use, + Add Load shows an alert until a column is removed or edited. • Any change to the load columns re-renders the engine table and recalculates all results.	

6. Engine Data Table

This is the main data-entry table. Each row is either a section header, a measured/default input row, a calculated correction-percentage row, or the final ISO-corrected result row. Columns run across the current engine load points.

ENGINE NO. 1 ORDER NO. optional		☐ Input fields ☒ Default values (can be changed) ☐ Calculation steps ☐ Result				
PARAMETER	ENGINE LOAD				UNIT	DESCRIPTION
	25%	50%	75%	100%		
SFOC MEASURED (FROM SHOP / SEA TRIAL TEST)						
SFOC Measured					g/kWh	Measured SFOC from test log
AMBIENT CONDITIONS ON SITE (ACTUAL)						
Tx					°C	Room / ambient air temperature at TC inlet
Px	1010.3	1010.3	1010.3	1010.3	mbar	Ambient total barometric pressure
Tcx					°C	Charge air coolant (jacket water) temperature
G (FO LCV)	42700	42700	42700	42700	kJ/kg	Net calorific value of fuel used
ISO CORRECTION FACTORS – LINEAR METHOD (PER TECHNICAL FILE)						
ΔT correction	–	–	–	–	%	= (Tx – 25) / 10 × 0.2% (0.2% per 10°C rise)
ΔP correction	–	–	–	–	%	= –(Px – 1000) / 10 × 0.02% (–0.02% per 10 mbar rise)
ΔTcx correction	–	–	–	–	%	= (Tcx – 25) / 10 × 0.6% (0.6% per 10°C rise)
ΔLCV correction	–	–	–	–	%	= –(G – 42700) × 100 / 42700 (–1% per 1% LCV rise)
Σ corrections	–	–	–	–	%	= ΔT + ΔP + ΔTcx + ΔLCV
SFOC ISO CORRECTED						
SFOC ISO Corrected	–	–	–	–	g/kWh	= SFOC_measured – (Σ% × SFOC_measured / 100)
Calculate Reset defaults						

Figure 4 — Engine Data table with sample shop-trial values and calculated ISO corrections.

6.1 Input and default fields

Field	Description	Unit	Type
SFOC Measured	Measured SFOC from the shop or sea trial test log.	g/kWh	Input
Tx	Room / ambient air temperature at the turbocharger inlet.	°C	Input
Px	Actual ambient total barometric pressure at the test site.	mbar	Default (editable)
Tcx	Charge-air coolant (jacket water) temperature.	°C	Input
G (FO LCV)	Net calorific value of the fuel used during the test.	kJ/kg	Default (editable)

6.2 Calculated correction fields

Field	Formula	Meaning
ΔT correction	$(Tx - Tr) / 10 \times 0.2\%$	0.2% correction per 10 °C difference from the ISO reference ambient temperature.
ΔP correction	$-(Px - pr) / 10 \times 0.02\%$	–0.02% correction per 10 mbar difference from the ISO reference pressure.
ΔTcx correction	$(Tcx - Tcr) / 10 \times 0.6\%$	0.6% correction per 10 °C difference from the ISO reference coolant temperature.

Field	Formula	Meaning
Δ LCV correction	$-(G - H_{ref}) \times 100 / H_{ref}$	Corrects for the difference between the actual fuel LCV and the ISO reference LCV.
Σ corrections	$\Delta T + \Delta P + \Delta T_{cx} + \Delta LCV$	Total percentage correction applied to the measured SFOC at that load point.

6.3 Result field

Field	Formula	Meaning
SFOC ISO Corrected	$SFOC_{measured} - (\Sigma \text{ corrections} \times SFOC_{measured} / 100)$	Final ISO 3046-corrected SFOC at each engine load point.

6.4 Row colour coding

Colour	Meaning
White	Direct measurement inputs entered by the user (SFOC Measured, Tx, Tcx).
Green	Editable default values that may be changed if the actual test record differs (Px, G).
Blue	Intermediate correction calculations generated automatically (ΔT , ΔP , ΔT_{cx} , ΔLCV , Σ).
Yellow / amber	Final ISO-corrected SFOC result row.

6.5 Table controls

Control	Function
Engine No. / Order No.	Free-text identification fields shown at the top of the table and included in the record.
Calculate	Runs the ISO correction across all load points and refreshes the Results Summary, charts and PME section.
Reset defaults	Restores the tool's built-in sample values for the current engine.

Tip

- Enter SFOC in g/kWh, temperatures in °C, pressure in mbar and fuel LCV in kJ/kg — the tool does not convert units automatically.
- Use values from calibrated shop trial or sea trial test records wherever possible; the corrected SFOC is only as reliable as the measured inputs.

7. Results Summary and Comparison Charts

After Calculate is clicked, the Results Summary and SFOC Comparison Charts sections appear below the engine table. Both update automatically whenever an input, a reference condition, or a load column is changed and Calculate is re-run.

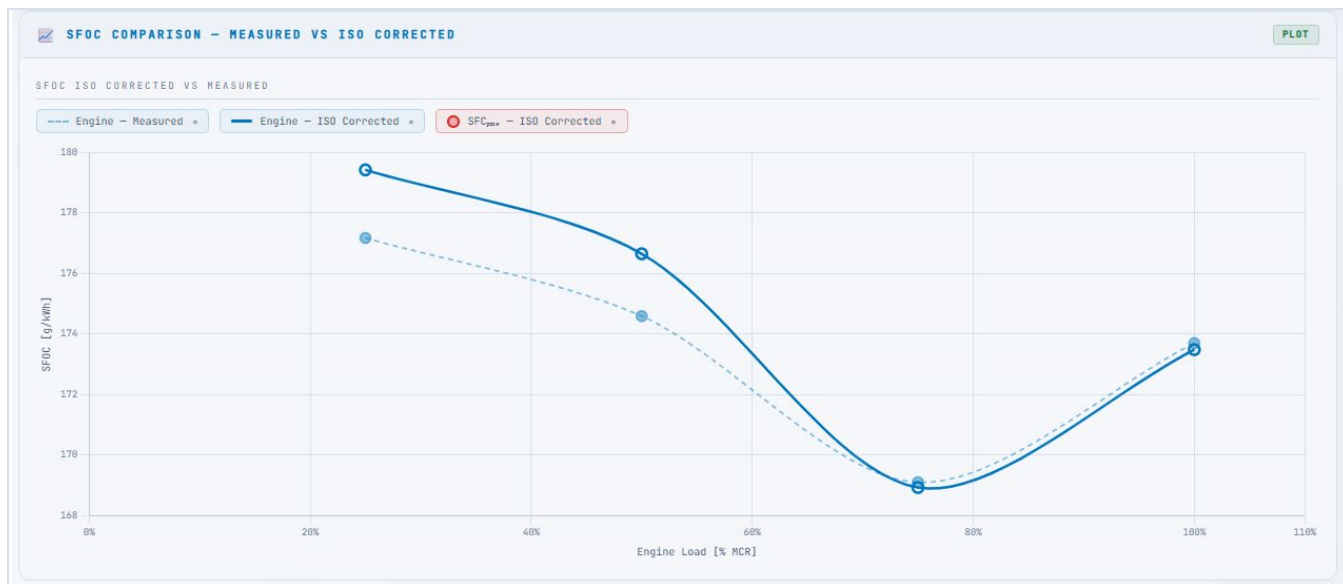


Figure 5 — SFOC ISO Corrected Results summary card.

The SFOC Comparison Charts section plots Measured SFOC against SFOC ISO Corrected for each load point, both as individual per-engine mini charts and as a combined overlay chart, so the effect of the ISO correction can be checked visually across the full load range before the figures are used for reporting or performance trending.

Tip

- If the Results Summary or charts still show placeholder values, click Calculate again — results do not update automatically while a field is being edited.

8. SFC Calculator for PME

This optional section estimates the ISO-corrected specific fuel consumption at the vessel's actual propulsion power (PME), using the ISO-corrected SFC curve generated from the engine table above as its reference.

SFC CALCULATOR FOR P_{ME}

Yes

No

Select Yes to show the P_{ME} calculator

Figure 6 — SFC Calculator for PME toggle (Yes / No).

Control	Function
Yes / No toggle	Switch Yes to open the SFC Calculator for PME panel; switch No to hide it.

Once switched on, the panel shows the reference curve, an MCRLim toggle, and the calculation result:

SFC CALCULATOR FOR P_{ME}

Yes

No

SFC Calculator for P_{ME} is shown

SFC CALCULATOR FOR P_{ME}

INTERPOLATED

ISO CORRECTED SFC — REFERENCE CURVE

FROM ENGINE TABLE

25% MCR

179.43

g/kWh

50% MCR

176.66

g/kWh

— LOWER BOUND

75% MCR

168.93

g/kWh

— UPPER BOUND

100% MCR

173.49

g/kWh

MCR_{LIM} AVAILABLE?

✓ YES

✗ NO

■ Using P_{ME} / MCR × 100 % Load

P_{ME} — PROPULSION POWER

350

kW · actual propulsion power

MCR — MAXIMUM CONTINUOUS RATING

510

kW · from engine nameplate

Formula (MCR_{LIM} = Yes):

Load% = P_{ME} / MCR × 100 → read SFC_{PME} from ISO Corrected SFC curve at that load% (Linear interpolation between the 25 / 50 / 75 / 100% MCR data points above)

SFC_{PME} — ISO CORRECTED

170.90

g / kWh

P_{ME} 350 kW ÷ MCR 510 kW = 68.6% MCR · SFC_{PME} Interpolated from ISO Corrected SFC curve

LOAD POINT

68.6%

METHOD

MCR_{LIM} = Yes

Figure 7 — SFC Calculator for PME, showing the ISO-corrected reference curve, PME/MCR inputs and the calculated SFC_{PME} result.

8.1 ISO Corrected SFC — Reference Curve

This row is populated automatically from the SFOC ISO Corrected values in the engine table above, shown at 25%, 50%, 75% and 100% MCR. It forms the curve used for the PME interpolation and cannot be edited directly.

8.2 MCRLim available? — Yes

Select Yes when an MCR limit value is available for the vessel. Enter PME (actual propulsion power, kW) and MCR (maximum continuous rating, kW). The tool calculates Load% = PME ÷ MCR × 100, then determines SFC_{PME} by linear interpolation between the two nearest ISO-corrected SFC load points.

8.3 MCRLim available? — No

Select No when no MCR limit value is available. In this mode no PME or MCR input is required — the tool uses the 75% MCR ISO-corrected SFC value directly as SFC_{PME}.

8.4 Result display and chart

The result panel shows SFCPME — ISO Corrected in g/kWh, the load point used, and the calculation method (interpolated or fixed 75% MCR). The SFCPME point is also plotted on the accompanying chart against the ISO-corrected SFC curve for a visual check.

Field	Unit	Type	Description
PME — Propulsion Power	kW	Input	Actual propulsion power used in the SFC calculator (only when MCRLim = Yes).
MCR — Maximum Continuous Rating	kW	Input	Value from the engine nameplate, used to determine the PME load percentage (only when MCRLim = Yes).
SFCPME	g/kWh	Result	ISO-corrected SFC at the selected PME operating point.

Tip

- The PME result is linked to the ISO-corrected engine table — any change to measured SFOC or the ISO reference conditions automatically updates the PME reference curve and result.
- Re-run Calculate on the engine table first if the reference curve still shows a dash (—).

9. Calculation Formulas Reference

The table below summarises every formula used by the calculator, per the vessel's linear-correction Technical File method.

Correction	Formula
Ambient temperature	$\Delta T = (T_x - T_r) / 10 \times 0.2\%$
Barometric pressure	$\Delta P = -(P_x - p_r) / 10 \times 0.02\%$
Coolant temperature	$\Delta T_{cx} = (T_{cx} - T_{cr}) / 10 \times 0.6\%$
Fuel LCV	$\Delta LCV = -(G - H_{ref}) \times 100 / H_{ref}$
Total correction	$\Sigma \text{ corrections} = \Delta T + \Delta P + \Delta T_{cx} + \Delta LCV$
SFOC ISO Corrected	$SFOC_{ISO} = SFOC_{measured} - (\Sigma \text{ corrections} \times SFOC_{measured} / 100)$
PME load percentage	$Load\% = PME / MCR \times 100$ (used when MCRLim = Yes)

10. Best-Practice Checklist

- Confirm the ISO Reference Conditions match the vessel's approved Technical File before entering test data.
- Use the same engine load points consistently across the table, the results and the PME reference curve.
- Enter SFOC in g/kWh, temperatures in °C, pressure in mbar and fuel LCV in kJ/kg — double-check units before calculating.
- Use calibrated shop trial or sea trial test records for all measured inputs wherever possible.
- Click Calculate after every change to inputs, reference conditions or load columns so the results, charts and PME section stay in sync.
- Review the SFOC Comparison chart for each load point before using the corrected figures for reporting or trending.
- Use Reset defaults only when you want to discard current entries and start again from the tool's sample values.

11. Troubleshooting

Symptom	Likely cause / fix
Correction fields show —	Calculate has not been run yet, or a required input field is empty — fill all input fields and click Calculate.
+ Add Load shows an alert	All four standard load columns (25%, 50%, 75%, 100%) are already in use — remove or edit an existing load chip first.
PME reference curve shows —	The engine table has not been calculated yet — click Calculate on the engine table before opening the PME calculator.
SFCPME does not update	Confirm the MCRLim toggle is set, and that PME and MCR values are entered (when MCRLim = Yes).
Chart looks empty	Allow a moment for the chart library to render, or click Calculate again after changing load points or inputs.
Reset defaults removed my data	Reset defaults restores the tool's built-in sample values for the current engine — re-enter your test data after using it.

Appendix A — Full Interface Reference

The overview below shows the complete calculator page with sample data entered, the ISO correction calculated, and the SFC Calculator for PME switched on, for reference alongside the section-by-section descriptions in this guide.

ISO 3046 REFERENCE T_2 AMBIENT 25 °C P_2 PRESSURE 1000 mbar T_{c2} COOLANT 25 °C H REF LCV 42700 kJ/kg

2 STROKE ENGINE Enter measured SFOC and ambient conditions — ISO correction applied automatically

25 % 50 % 75 % 100 % + Add Load Use + Add Load or x to update columns, summary and charts.

ENGINE NO. 1 ORDER NO. optional

Input fields Default values (can be changed) Calculation steps Result

PARAMETER	ENGINE LOAD				UNIT	DESCRIPTION
	25%	50%	75%	100%		
SFOC MEASURED (FROM SHOP / SEA TRIAL TEST)						
SFOC Measured					g/kWh	Measured SFOC from test log
AMBIENT CONDITIONS ON SITE (ACTUAL)						
T_x					°C	Room / ambient air temperature at TC inlet
P_x	1019.3	1019.3	1019.3	1019.3	mbar	Ambient total barometric pressure
T_{cx}					°C	Charge air coolant (jacket water) temperature
G (FO LCV)	42700	42700	42700	42700	kJ/kg	Net calorific value of fuel used
ISO CORRECTION FACTORS — LINEAR METHOD (PER TECHNICAL FILE)						
ΔT correction	—	—	—	—	%	$= (T_x - 25) / 10 \times 0.2\%$ (0.2% per 10°C rise)
ΔP correction	—	—	—	—	%	$= -(P_x - 1000) / 10 \times 0.02\%$ (-0.02% per 10 mbar rise)
ΔT_{cx} correction	—	—	—	—	%	$= (T_{cx} - 25) / 10 \times 0.6\%$ (0.6% per 10°C rise)
ΔLCV correction	—	—	—	—	%	$= -(G - 42700) / 100 \times 1\%$ (-1% per 1% LCV rise)
Z corrections	—	—	—	—	%	$= \Delta T + \Delta P + \Delta T_{cx} + \Delta LCV$
SFOC ISO CORRECTED						
SFOC ISO Corrected	—	—	—	—	g/kWh	$= SFOC_{measured} - (Z\% \times SFOC_{measured} / 100)$

Calculate Reset defaults

SFOC ISO CORRECTED — RESULTS
ISO 3046 corrected SFOC across all load points

SFOC COMPARISON — MEASURED VS ISO CORRECTED

SFOC ISO CORRECTED VS MEASURED

2 STROKE ENGINE — ISO SFOC CORRECTION CALCULATOR

SFC CALCULATOR FOR P_{ME} Yes No SFC Calculator for P_{ME} is shown

SFC CALCULATOR FOR P_{ME} INTERPOLATED

ISO CORRECTED SFC — REFERENCE CURVE FROM ENGINE TABLE

25% MCR 50% MCR 75% MCR 100% MCR

MCR₁₅₀ AVAILABLE ☒ YES ☐ NO Select MCR₁₅₀ option above

P_{ME} — PROPULSION POWER 350 kW — actual propulsion power

MCR — MAXIMUM CONTINUOUS RATING 510 kW — from engine nameplate

Method: Select MCR₁₅₀ option to see the applicable formula.

Figure 8 — Full calculator page after data entry, ISO calculation and PME calculation.

End of user guide.